

Irregular Shape Identifies Ruptured Intracranial Aneurysm in Subarachnoid Hemorrhage Patients With Multiple Aneurysms

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Background and Purpose—We investigated which aneurysm-related risk factors for rupture best discriminate ruptured versus unruptured saccular intracranial aneurysms (sIAs) in subarachnoid hemorrhage patients with multiple sIAs.

Methods—We included 264 subarachnoid hemorrhage patients with a ruptured sIA and at least one additional unruptured sIA, from the Kuopio Intracranial Aneurysm database from 2003 to 2015. These patients had 268 ruptured and 445 unruptured sIAs. Angiograms of the 713 sIAs were reevaluated for multiple variables describing aneurysm shape. Multivariate generalized linear mixed models were used to calculate odds ratios with corresponding 95% confidence intervals for the independent risk factors for aneurysm rupture.

Results—In the multivariate analysis, only sIA size ($P<0.004$) and irregular shape ($P<0.000$) independently associated with sIA rupture. As an independent risk factor, irregular shape showed the strongest association with rupture (odds ratio 90.3; 95% confidence interval, 47.0–173.5). The sIA location, flow angles, bottleneck factor, or aspect ratio were not significantly associated with rupture.

Conclusions—Irregular shape may identify the ruptured sIA better than size in patients presenting with aSAH and multiple sIAs. (*Stroke*. 2017;48:1986–1989. DOI: 10.1161/STROKEAHA.117.017147.)

Key Words: Finland ■ intracranial aneurysm ■ microsurgery ■ risk factors ■ stroke ■ subarachnoid hemorrhage

Rupture of a saccular intracranial aneurysm (sIA) causes aneurysmal subarachnoid hemorrhage (aSAH) with high mortality even with modern neurointensive care in the acute phase. Known risk factors for SAH include the size, shape, and location of the aneurysm.¹

Multiple sIAs are present in 15% to 30% of sIA carriers in recent patient series^{1–8} (Table I in the [online-only Data Supplement](#)). Because endovascular treatment has become the treatment choice for most ruptured sIAs, it is important to identify correctly the ruptured IA because unlike in microsurgery, in endovascular treatment, one cannot visually confirm rupture.

We investigated which aneurysm-related factor best identifies a ruptured sIA in 264 Eastern Finnish aSAH patients with 713 multiple sIAs admitted to Kuopio University Hospital (KUH) from 2003 to 2015.

Materials and Methods

Basic Study Population and Kuopio Intracranial Aneurysm Database

Study population included all (n=845) SAH patients admitted to KUH from its catchment area in Eastern Finland from 2003 to 2015. Database of all sIA cases has been collected in KUH since 1977.⁹

Diagnosis of sIAs

All patients with a suspicion of SAH routinely had a head computed tomography and, if negative, a lumbar puncture. SAH patients routinely had undergone computed tomographic angiography and digital subtraction angiography with 3-dimensional reconstructions. The ruptured sIA was identified based on the distribution of blood in computed tomography in cases that were treated endovascularly. The ruptured sIA was further confirmed during microsurgery in 150 patients with 233 aneurysms.

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Study Population

The final study population consisted of 264 aSAH patients with 713 sIAs fulfilling the following criteria (Figure 1):

- Admitted alive to KUH at the time of the first aSAH between 2003 and 2015;
- At least 2 saccular intracranial aneurysms verified in 4-vessel angiography;
- Angiogram (2-dimensional digital subtraction angiography with multiple projections, 3-dimensional digital subtraction angiography, magnetic resonance angiography, or computed tomographic angiography) images available for review.

Morphology of sIAs

All angiograms of the 713 sIAs were reevaluated and measured by 2 authors (J.B. and A.L.) on a 0.1-mm scale (Figure 2). Aspect ratios (dome height/neck width) and bottleneck factors (dome width/neck width) were calculated. Aneurysms were categorized as smooth or irregular.¹⁰ Aneurysms with multiple lobes, daughter sacs, or other types of wall protrusions were considered irregular. Inter-observer variance was calculated with Fleiss κ . Angles were measured on a 1-degree scale. For further analysis, angles were divided into tertiles.

Statistical Analysis

IBM SPSS Statistics 23 (IBM Corp, Armonk, NY) was used. To control for correlation between sIAs within the same patient with shared risk factors, we used logistic mixed model with patient as a random effect. We calculated odds ratios (ORs) with 95% confidence intervals (CIs) for the independent risk factors for aneurysm rupture. P values <0.05 were considered significant. The ORs were adjusted for patient-related risk factors (age, sex, smoking, hypertension, and familial history of aneurysms). The specificity and sensitivity of aneurysm characteristics as indicators of sIA rupture were investigated by receiver operating characteristic curves.

Ethical Aspects

The ethics committee of KUH approved the study. Informed consent was obtained from all patients.

Results

Study Population

Altogether, 713 sIAs (268 ruptured and 445 unruptured) were found in the 264 aSAH patients with multiple sIAs (Table 1; Table II in the [online-only Data Supplement](#)). Four patients had bled twice from separate sIAs.

Ruptured and Unruptured sIAs

The median sizes for the 268 ruptured and 445 unruptured sIAs were 6.9 and 2.5 mm, respectively (Table 1). Of the ruptured sIAs, 53.7% were smaller than 7 mm and 8.2% were smaller than 3 mm. Of the ruptured sIAs, 96.3% were irregularly shaped, in contrast to 8.1% of the unruptured ones. The largest sIA was not the one that ruptured (median 7.0 mm) in 33 patients (12.5%). In 29 of those patients, the largest aneurysm was smooth. However, in all 33 of these patients, the ruptured sIA was irregular.

Independent Risk Factors for sIA Rupture

In the multivariate analysis, sIA size (OR, 90.3; 95% confidence interval [CI], 47.0–173.5; $P<0.004$) and shape (OR, 1.20 per 1 mm; 95% CI, 1.06–1.37; $P<0.000$) independently associated with sIA rupture.

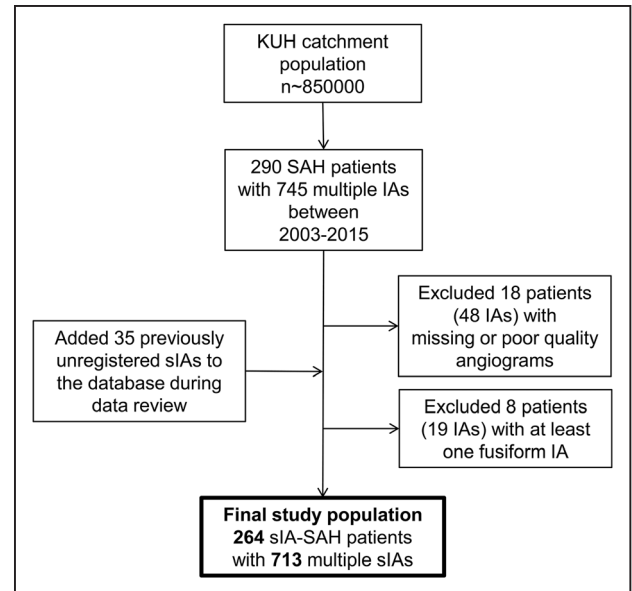


Figure 1. Flow chart showing the final study population of 264 subarachnoid hemorrhage (SAH) patients with 713 multiple saccular intracranial aneurysms (sIAs).

Subanalysis of Microsurgically Treated Patients

We performed sensitivity analysis to explore potential bias because of misclassification of sIA rupture status by subanalysis of microsurgically treated patients ($n=150$) with 100 unruptured and 133 ruptured sIAs. In the logistic mixed model, only sIA shape (OR, 43.2; 95% CI, 15.6–119.9) and size (OR, 1.24; 95% CI, 1.01–1.54) showed significant association with rupture status.

Diagnostic Accuracy of the Aneurysm Characteristics as Indicators of sIA Rupture

We tested the diagnostic accuracy of predictors of sIA rupture by plotting receiver operating characteristic curves (Figure I in the [online-only Data Supplement](#)). The sIA shape and size had the best diagnostic value with areas under the curve of 0.941 and 0.862, respectively.

Discussion

The ruptured sIA was not the largest in 13% in our study cohort of 264 aSAH patients with 713 multiple sIAs, from a prospectively collected, population-based sIA database. Of the 268 ruptured sIAs, 96.3% were irregularly shaped in contrast to 8.1% of the 445 unruptured sIAs. In multivariate analysis, irregular shape showed the strongest association

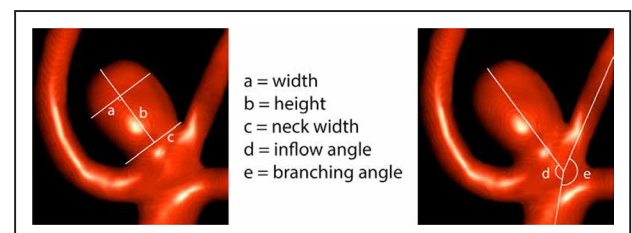


Figure 2. Measurements of the aneurysm: dimensions (left) and related angles (right).

Table. Characteristics of the 268 Ruptured and 445 Unruptured sIAs

	Ruptured	Unruptured	<i>P</i> Value	Exp	95% CI for OR	
	n=268 (37.6%)	n=445 (62.4%)			Upper	Lower
Aneurysm location						
Internal carotid artery*	26 (9.7%)	104 (23.4%)	Ref	...	...	...
Posterior communicating artery	40 (14.9%)	31 (7.0%)	0.980	0.985	0.306	3.175
Anterior cerebral artery†	16 (6.0%)	32 (7.2%)	0.397	1.829	0.452	7.403
Anterior communicating artery	67 (25.0%)	45 (10.1%)	0.194	2.022	0.699	5.851
Middle cerebral artery	89 (33.2%)	197 (44.3%)	0.385	0.666	0.266	1.666
Posterior circulation	30 (11.2%)	36 (8.1%)	0.268	1.982	0.589	6.666
Aneurysm size, mm						
Median	6.9	2.5	0.004	1.203	1.060	1.367
Aspect ratio, median	1.8	1.0	0.974	0.989	0.501	1.949
Bottleneck factor, median	1.4	1.0	0.507	1.319	0.581	2.991
Aneurysm shape						
Smooth	10 (3.7%)	409 (91.9)	Ref	...	...	...
Irregular	258 (96.3%)	36 (8.1%)	0.000	88.343	45.903	170.020
Flow angles, °						
Inflow angle						
55–100	19 (7.1%)	75 (16.9%)	Ref	...	...	...
100–140	119 (44.4%)	232 (52.1%)	0.847	0.911	0.351	2.360
140–180	130 (48.5%)	138 (31.0%)	0.761	0.854	0.308	2.369
Branching angle						
55–100	34 (12.7%)	26 (5.8%)	0.469	0.664	0.219	2.017
100–140	145 (54.1%)	196 (44.0%)	0.793	0.918	0.486	1.734
140–180	89 (33.2%)	223 (50.1%)	Ref	...	...	...

CI indicates confidence interval; OR, odds ratio; Ref, reference; and sIA, saccular intracranial aneurysm.

*Other than posterior communicating artery.

†Other than anterior communicating artery.

with sIA rupture after adjustment for aneurysm- and patient-related risk factors.

Our Findings Versus Previous Data

In a Japanese follow-up study of >5000 patients, sIA size, location, and the presence of a daughter sac were associated with sIA rupture.² Furthermore, in a series of 124 aSAH patients with 302 multiple sIAs from the Netherlands, size, aspect ratio, and aneurysm shape associated with rupture.³ In this study, irregular shape strongly associated with sIA rupture, independent of size, location, and patient-related factors. The sIAs located in anterior communicating artery and posterior communicating artery ruptured more often than sIAs in other locations (Table), a phenomenon seen in previous SAH cohorts.^{1,2,7,9,10}

Strengths of the Present Study

Strength of this study is the large population-based cohort of aSAH patients with high-quality angiograms. In multivariate analysis, we adjusted not only for known risk factors of aneurysm rupture but also for patient-related factors.

Limitations of the Study

There is a chance of misclassification of the rupture status of sIAs, especially in cases where multiple sIAs are located in the same artery close to the maximum hemorrhage intensity. To control for that, we performed sensitivity analysis with microsurgically treated patients with 233 aneurysms, in which the rupture status was visually confirmed by the operating vascular neurosurgeon. This was in line with our main analysis because only sIA shape and size were statistically significantly associated with sIA rupture. Irregular shape had the strongest association with sIA rupture.

Clinical Applications

Because of development of the imaging modalities during the study period, there is inherent variation in the image quality, especially in the 3-dimensional reformations acquired from the aneurysms. This may have led to overinterpretation of the irregularity of the aneurysm wall, especially in the early 3-dimensional surface reconstructions. However, the single-center configuration of the study compensates for this because all

of the aneurysms were imaged with consistent guidelines and techniques by a dedicated group of radiologists. The interrater reliability was tested in randomly selected set of 50 patients with 137 sIAs. The Fleiss κ value for 5 independent raters was 0.68, showing that level of agreement was substantially high.

Gadolinium-enhanced magnetic resonance imaging can be used to identify a ruptured intracranial aneurysm in aSAH patients with multiple IAs.¹¹ However, visually assessed shape of aneurysms is robust tool to locate the ruptured IA, requires no additional imaging, and is easy to use in all clinical settings.

Conclusions

Irregular shape may identify the ruptured sIA better than size in patients presenting with aSAH and multiple sIAs.

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Disclosures

None.

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